What is the difference between the `Framework` and the `Library` terms?

Both frameworks and libraries are code written by someone else that is used to help solve common problems.

FrameWork

A framework is like the foundation upon which developers build applications for specific platforms. It includes reusable pieces of code written to perform common tasks and uses code provided by a developer for custom functionality.

- Frameworks combine resources like `image` `files` and reference `documents` into one package. It is thus possible to modify that package to fit specific project requirements. With a framework, developers can integrate new features into an application and give it new capabilities.

Why do we use Frameworks?

Software development is a complex process. This involves a lot of phrases like coding, designing, testing, etc. When it comes to coding, developers must worry about syntax, declarations, garbage collection, statements, exceptions, and more. Software frameworks make development easier by providing a common platform from which developers can control all or most of the software development process.

Library

A library is a collection of prewritten code that can be used to simplify tasks. The term library simply refers to a collection of code that is reused repeatedly.

- Usually, most libraries do not have a large size. Generally, libraries are focused on narrow scopes such as `strings`, `sockets`, and `IO`, etc., so their `APIs` are also small and require fewer dependencies. `jQuery`, `NumPy`, etc. are examples of libraries.

Why do we need a library?

There is only one answer to this question, and that is to reuse the code that has already been written by someone else (or another developer). Developers can avoid writing code for functionality that is already written in the library by using it. The result is more efficiency and less time spent on coding. Since it's more likely that other people are using it too, you'll benefit from them finding and fixing any bugs. This is one of the reasons libraries are useful.

Key Differences

1. In some ways, you can think of a framework as a collection of libraries, but it's entirely different. By using a library, you control the flow of the program. The library can be invoked whenever and wherever you like. Contrary to this, when you use a framework, the flow is controlled by the framework. The framework instructs you where to put your code, but it will call your code as required. Simply put, our code calls the library's code, but in a framework, it's the framework's code that calls our code as shown in the below diagram.

2. Developers can invoke libraries to perform specific tasks by using components, classes, and methods. A framework, however, already provides code to perform common tasks and uses code provided by a developer for custom functionality.

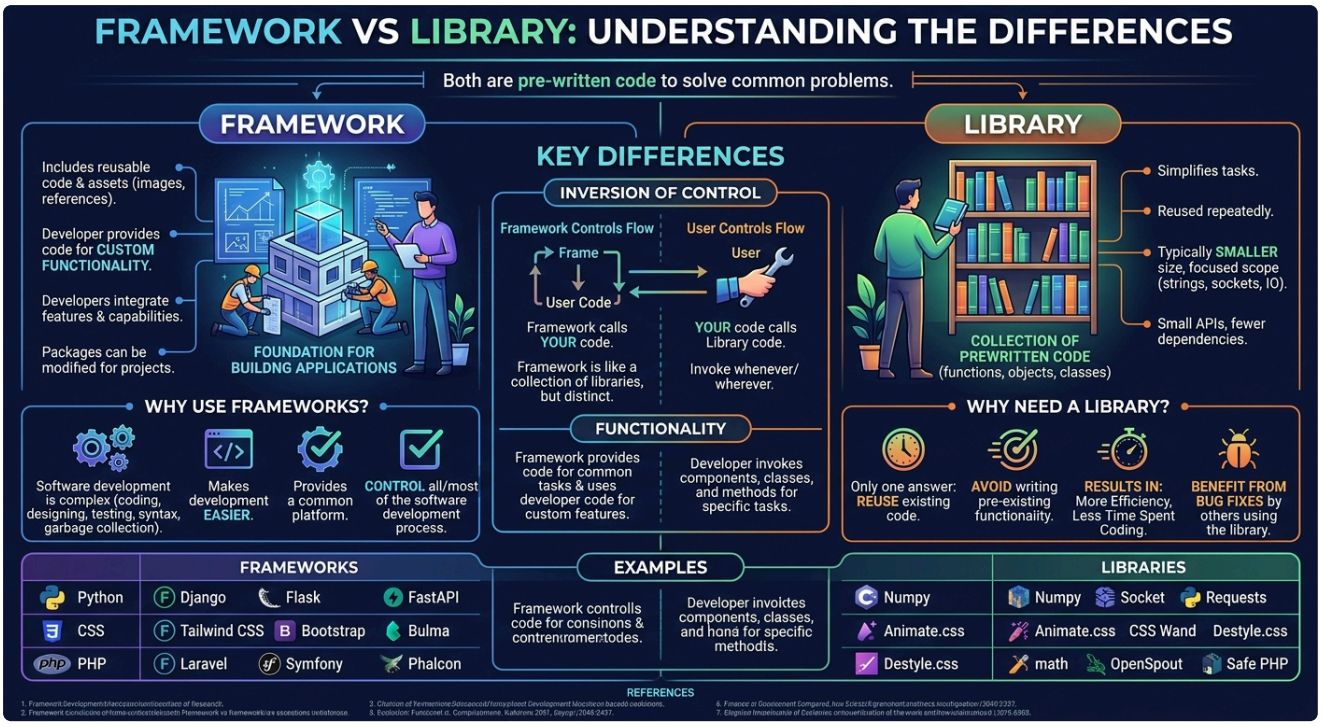
Examples

	Framworks	Libraries
Python	1. Dango 2. Flask 3. FastAPI	1. Numpy 2. Socket 3. Requists
Css	1. Tailwind CSS 2. Bootstrap 3. Bulma	1. Animate.css 2. CSS Wand 3. Destyle.css

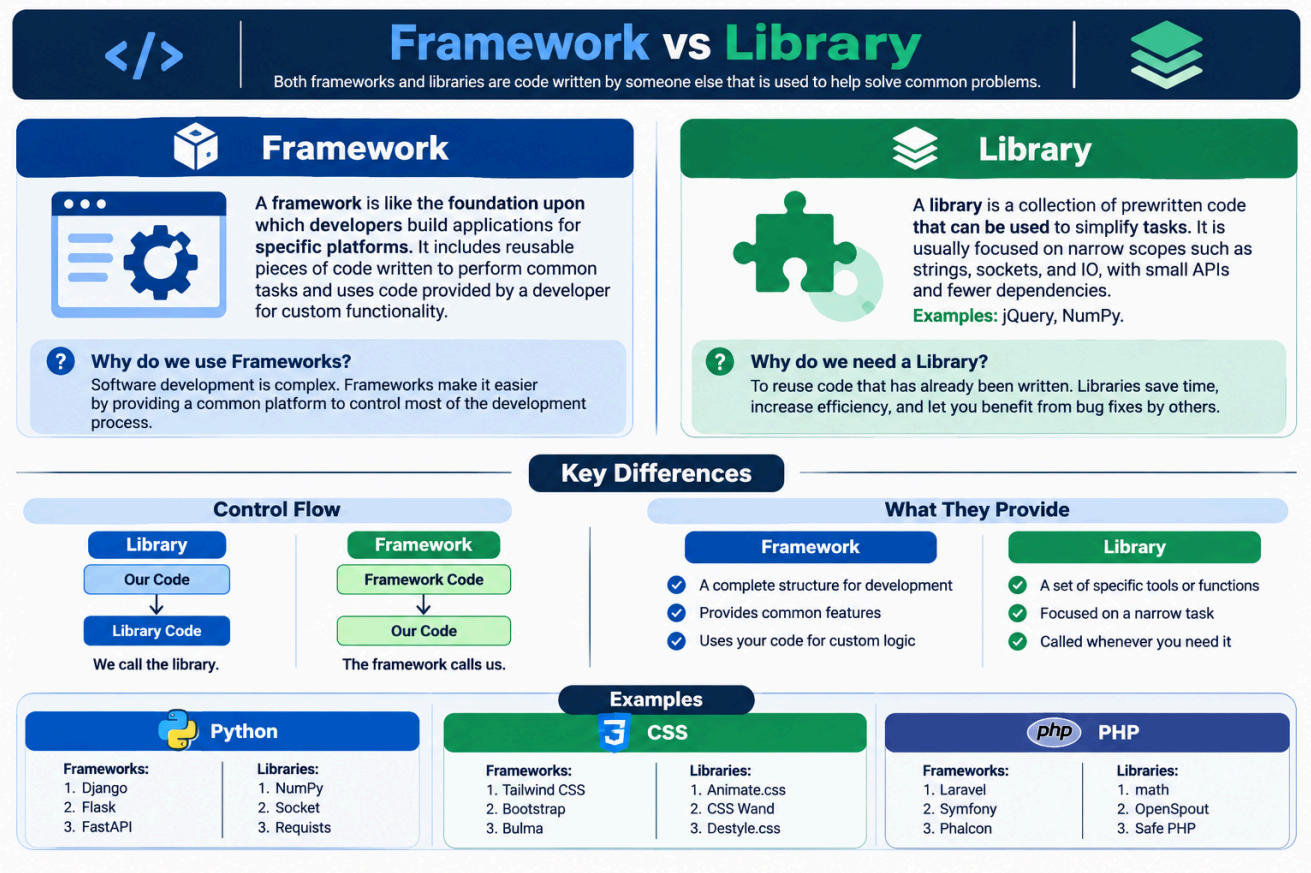
	Framworks	Libraries
PHP	1. Laravel 2. Symfony 3. Phalcon	1. math 2. OpenSpout 3. Safe PHP

Info Graph

`Gemini Auto:` free edition



`ChatGPT` Free edition



References

	Research Link
1	https://www.interviewbit.com/blog/framework-vs-library/
2	https://www.reply.com/valorem-reply/en/resources/insights/blog/2025/april/6-best-css-frameworks-for-developers-in-2026
3	https://tiffanylewis.hashnode.dev/best-20-css-libraries-for-developers-and-designers-2024
4	https://www.geeksforgeeks.org/blogs/php-frameworks/
5	https://dev.to/rubenrubiob/php-libraries-and-tools-4bk3